

Radionuclide Identification for Plastic Scintillation Detectors by Backward Cumulative Channel Sum

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Abstract

Large-area plastic scintillation detectors are widely used in radiation portal monitors, but their Compton-dominated spectra provide limited radionuclide-specific information. We evaluated a backward cumulative channel sum (BCCS) representation that integrates counts above successive channel thresholds and combines the resulting profiles with non-negative least-squares (NNLS) decomposition. Experimental 20-s spectra from a large-area plastic-scintillator radiation portal monitor were analyzed using background-subtracted BCCS-NNLS. Background-normalized profiles were used only for visualization. Under a known-constituent-number evaluation, both BCCS-NNLS and a detector-specific five-window NNLS baseline ranked the expected constituent pair in the top two across eight binary mixed-source spectra. In the ternary mixture, NNLS returned positive coefficients for all three available reference components; because the candidate library contained only the three radionuclides, this result was interpreted as decomposition consistency rather than a discriminative top-K identification test. Relative residuals calculated within the respective representation spaces ranged from 0.37% to 1.24% for BCCS-NNLS and from 0.77% to 2.55% for window-NNLS. Energy-weighted spectra provided qualitative corroboration of spectral features but were sensitive to high-channel statistical fluctuations and suppression of weak Am-241 responses at higher weighting orders. BCCS is linear and peak-independent and retains a continuous sequence of cumulative spectral thresholds, enabling mixed-source decomposition without a training dataset. Because the cumulative coordinates are strongly correlated and the present comparison includes limited independent repetitions, these results demonstrate the practical feasibility of BCCS-NNLS rather than its general performance superiority.

Keywords: plastic scintillator; radiation portal monitor; radionuclide identification; backward cumulative channel sum; non-negative least squares; energy window; energy weighting

Introduction

Radiation portal monitors (RPMs) are deployed at border crossings, ports, transportation facilities, and industrial sites to detect unauthorized movement of radioactive material. Large polyvinyl toluene or related plastic scintillation panels are attractive for this purpose because they provide large sensitive areas, fast response, mechanical robustness, and moderate cost [1,2]. Their principal spectrometric limitation is poor energy resolution: the recorded pulse-height distribution is dominated by broad Compton continua rather than narrow full-energy peaks. Radionuclide identification must therefore use distributed spectral shape information while operating against a background that may be larger than the net source response.

Several low-dimensional and data-driven approaches have been proposed. Energy-window algorithms divide the spectrum into a small number of regions and compare integrated counts or ratios. They are computationally simple and have been deployed in RPM applications, but the result depends on detector- and gain-specific window boundaries and the within-window shape is discarded [3,4]. Energy-weighted methods multiply channel counts by progressively stronger energy terms to transform broad Compton features into detectable maxima. These methods can provide useful peak-like signatures, although higher-order weighting is nonlinear and can suppress low-energy responses [5,6]. Principal component, neural-network, and convolutional approaches can exploit more complex patterns, but normally require representative training data and validation across operating environments [7-11].

The present work examines a simpler representation: the Backward Cumulative Channel Sum (BCCS). At each channel, BCCS reports the count integrated from that threshold to the highest acquired channel. The operation visually reduces isolated channel-to-channel fluctuations while preserving a continuous sequence of cumulative spectral thresholds. More importantly, it is a linear transformation. The BCCS of a mixed spectrum equals the sum of the constituent BCCS spectra, making the representation compatible with NNLS decomposition [12].

Accordingly, this study evaluates whether background-subtracted BCCS combined with NNLS can decompose weak and mixed radionuclide responses acquired with a plastic-scintillator RPM. The results are compared with raw-spectrum NNLS, a detector-specific energy-window approach, and a published energy-weighted method, and the weak-signal ranking stability was examined using distance-dependent Cs-137 measurements.

Results

Reference spectra and BCCS transformation

Figure 1 compares background and reference spectra for Am-241, Cs-137, and Co-60. The raw plastic-scintillator spectra contain broad continua, with the Am-241 response concentrated near the lower-level discriminator (LLD). After backward cumulative transformation, each point represents the integrated count above a successive channel threshold. The curves are smooth and monotonically decreasing, but neighboring BCCS points are not statistically independent because they share most of the same counts. Background-subtracted BCCS was used for all primary quantitative analyses, whereas the background-ratio representation defined in Methods is shown only for visualization.

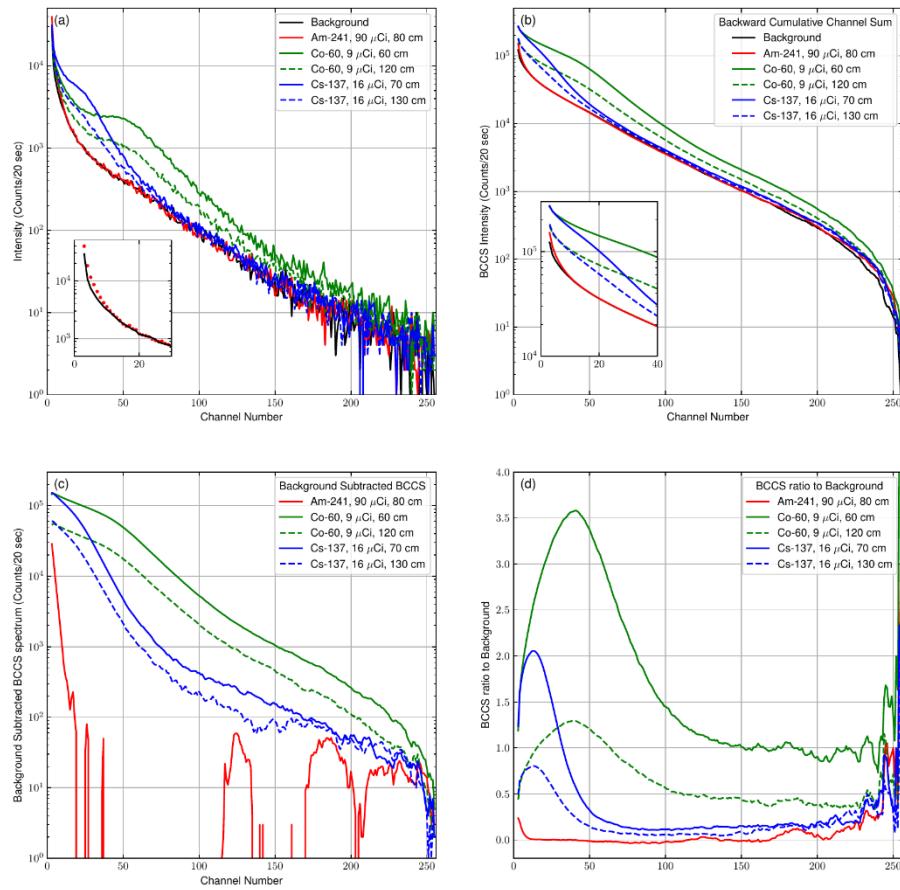


Figure 1. Reference spectra before and after BCCS processing. (a) Raw 20-s spectra of background, Am-241, Cs-137, and Co-60. (b) Corresponding backward cumulative spectra. (c) Background-subtracted BCCS, which preserves net count scale. (d) BCCS ratio to background, retained as a secondary visualization. The apparent smoothness of the cumulative curves should not be interpreted as statistical independence of adjacent thresholds.

Mixed-source spectral structure

The nine mixed-source measurements showed source-dependent differences in their background-subtracted cumulative profiles (Figure 2). Differences were most pronounced over the lower cumulative threshold coordinates, although each coordinate included counts extending to Channel 256. The background-ratio representation provided a complementary visualization but was not used for quantitative decomposition.

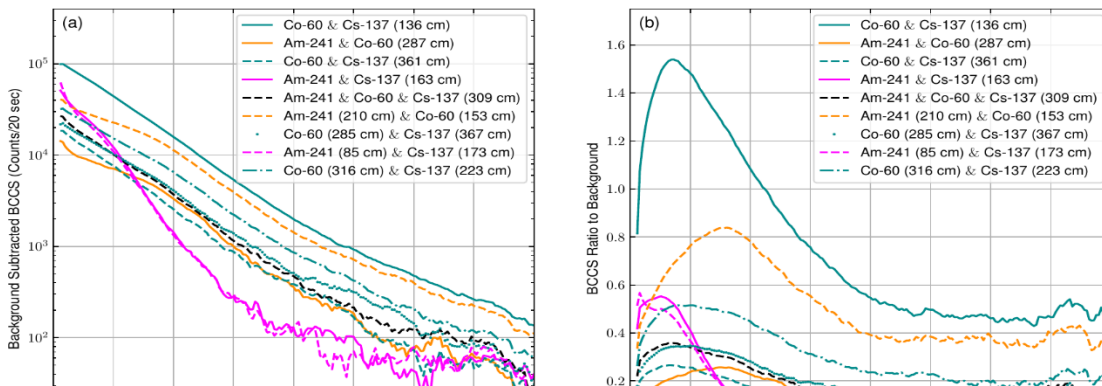


Figure 2. Processed BCCS spectra for the nine mixed-source measurements included in the submitted dataset. (a) Background-subtracted BCCS retains absolute net count-rate differences. (b) The BCCS ratio to background emphasizes relative shape differences. All nine mixed-source spectra were analyzed using BCCS-NNLS and five-window NNLS. Samples 13 and 20 were selected for detailed visualization because they represent intermediate-to-high-energy and low-to-intermediate-energy binary mixtures, respectively.

Quantitative BCCS-NNLS decomposition

Sample-13 contained Cs-137 and Co-60. BCCS-NNLS returned relative reference scaling coefficients of 0.951 for Cs-137 and 0.796 for Co-60; the Am-241 coefficient was 0.031. The relative reconstruction residual was 0.37%. Sample-20 contained Am-241 and Cs-137. The corresponding coefficients were 0.946 for Am-241 and 0.610 for Cs-137; the Co-60 coefficient was 0.018. The relative residual was 0.77%. The small non-constituent coefficients are reported rather than silently set to zero because they indicate reference mismatch and the practical need for a validated multi-label decision rule.

Raw-spectrum NNLS with the measured background included as an additional template produced the same constituent rankings for Samples 13 and 20, with representation-specific residuals of 5.02% and 3.76%, respectively. The numerical BCCS residuals were smaller than the raw-spectrum residuals, but this difference is descriptive only because the residuals were calculated in different representation spaces and the cumulative coordinates have non-diagonal covariance.

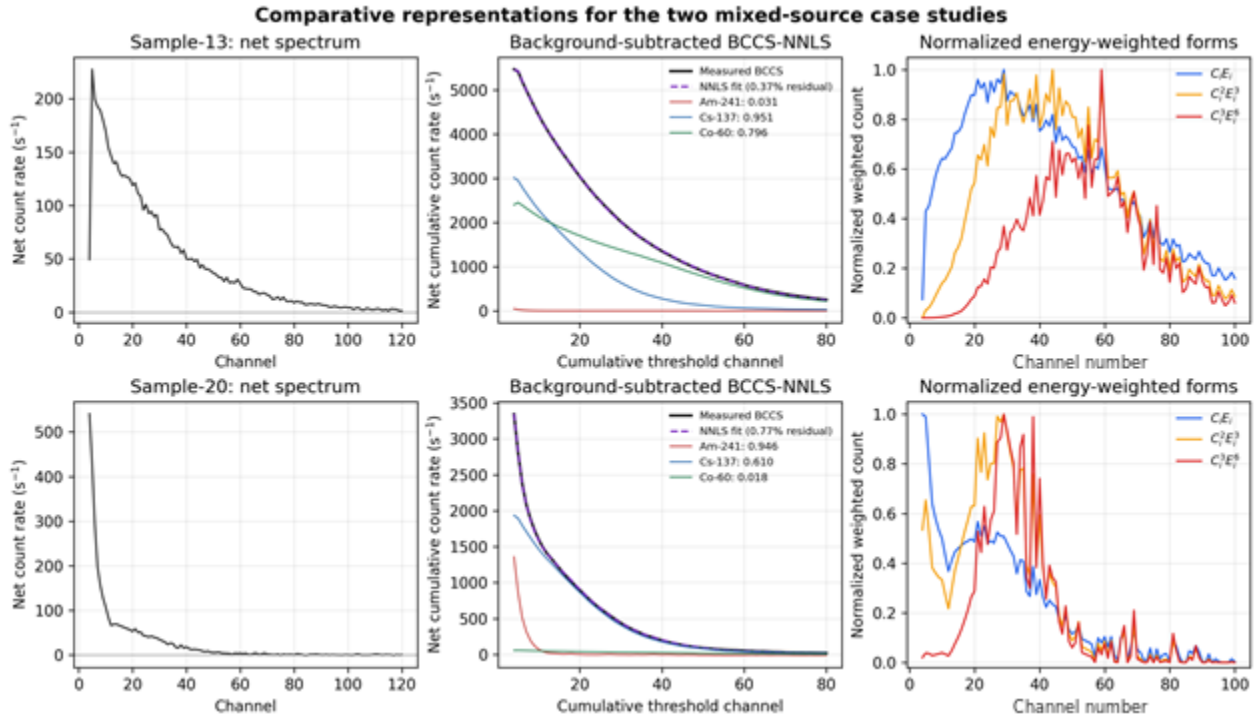


Figure 3. Comparative representations for two mixed-source case studies. The left column shows the background-subtracted channel spectra. The center column shows measured and reconstructed BCCS profiles with NNLS component curves. The right column shows the three normalized energy-weighted representations. Sample-13 contained Cs-137 and Co-60; Sample-20 contained Am-241 and Cs-137. Energy weighting uses a channel-proportional energy coordinate because identification intervals were derived from the same detector-specific reference spectra.

Comparison with energy-window and energy-weighted methods

A detector-specific five-window baseline compressed the 253 active channels into five intervals: Ch. 4–10, 11–25, 26–40, 41–80, and 81–256. NNLS decomposition of the five integrated responses, with the measured background included as an additional window template, ranked the expected constituent pair as the two largest radionuclide coefficients for both representative binary mixtures. This representation is linear and computationally efficient but discards spectral variation within each interval. The relative residuals were 1.05% for Sample-13 and 0.77% for Sample-20.

As an adapted qualitative implementation of the energy-weighted method reported by Lee et al. [6], three normalized transformations $C_i E_i$, $C_i^2 E_i^3$, and $C_i^3 E_i^6$ were evaluated. A radionuclide was qualitatively assigned when a reference-consistent feature was observed in at least two of the three representations. This operational criterion was used for qualitative corroboration and was not treated as a validated open-set classification rule. Sample-13 exhibited Cs-137-consistent features in the first two representations and Co-60-consistent features in the second and third representations. Sample-20 exhibited Am-241-consistent features in the first two representations and Cs-137-consistent features in all three. The resulting qualitative assignments agreed with those obtained using BCCS-NNLS and five-window NNLS. However, increasing the weighting order progressively reduced the relative prominence of the low-channel Am-241 response.

Across the eight binary mixed-source spectra, the expected constituent radionuclides occupied the two largest coefficients for both BCCS-NNLS and five-window NNLS. In the ternary Sample-17 spectrum, both methods returned positive coefficients for all three available reference components. Because the candidate

library contained only these three radionuclides, this result was interpreted as decomposition consistency rather than discriminative top-K identification. BCCS-NNLS returned coefficients of 0.171 for Am-241, 0.225 for Cs-137, and 0.187 for Co-60, with a relative residual of 1.24%; the corresponding five-window residual was 2.55%. Across all nine mixtures, the relative residuals ranged from 0.37% to 1.24% for BCCS-NNLS and from 0.77% to 2.55% for five-window NNLS. Because these residuals were calculated in their respective representation spaces, their numerical magnitudes should not be interpreted as directly comparable measures of method performance. These results represent known-constituent-number rankings and should not be interpreted as open-set multi-label identification accuracy.

Table 1. NNLS decomposition of two representative binary mixed-source spectra using the same set of single-radionuclide reference spectra.

Sample	Representation	Am-241	Cs-137	Co-60	Residual	Expected /Top-2 assignment
13	Raw-spectrum NNLS	0.047	0.966	0.747	5.02%	Cs-137 + Co-60
	BCCS-NNLS	0.031	0.951	0.796	0.37%	Cs-137 + Co-60
	5-window NNLS	0.000	0.964	0.776	1.05%	Cs-137 + Co-60
20	Raw-spectrum NNLS	0.932	0.624	0.000	3.76%	Am-241 + Cs-137
	BCCS-NNLS	0.946	0.610	0.018	0.77%	Am-241 + Cs-137
	5-window NNLS	0.941	0.617	0.000	0.77%	Am-241 + Cs-137

Raw-spectrum NNLS with a measured background template was additionally applied to all nine mixed-source measurements. The fitted radionuclide and background coefficients, original-channel-space residuals, and known-K rankings are summarized in Supplementary Table S3. Among the eight binary mixtures, the expected constituent pair was included among the two largest radionuclide coefficients in all eight cases. The ternary Sample-17 result was not evaluated using the top-K metric because the number of expected constituents equaled the number of candidate radionuclides.

The coefficients in Table 1 are detector-response scaling factors relative to the corresponding single-radionuclide reference measurements; they do not represent radionuclide activity fractions. Because the reference and mixed-source measurements differ in source activity, source-to-detector distance, geometry, and energy-dependent detector response, the coefficient ratios cannot be converted into activity fractions without an appropriate efficiency and geometry model.

Table 2. Qualitative identification using the three 2020 energy-weighted representations.

Sample	$C_i E_i$	$C_i^2 E_i^3$	$C_i^3 E_i^6$	Assignment
13	Cs-137-consistent feature near Ch. 27	Cs-137 and Co-60-consistent features near Ch. 31 and 57	Co-60-consistent feature near Ch. 57	Cs-137 + Co-60
20	Am-241 and Cs-137-consistent features near Ch. 6 and 21	Am-241 and Cs-137-consistent features near Ch. 6 and 29	Cs-137-consistent feature near Ch. 29	Am-241 + Cs-137

Features were assigned by comparison with the corresponding transformed single-radionuclide reference spectra. A radionuclide was qualitatively assigned when a reference-consistent feature was observed in at least two representations. This criterion was used only for qualitative comparison and does not constitute a validated open-set classification procedure.

Distance-dependent Cs-137 evaluation

Cs-137 was measured at 70, 130, 300, and 390 cm. The background-subtracted count rate decreased from 8066.8 s⁻¹ at 70 cm to 460.1 s⁻¹ at 390 cm. Using the 130-cm spectrum as the fixed Cs-137 reference, the

BCCS-NNLS coefficient decreased from 2.581 to 0.128. Cs-137 remained the largest coefficient at every distance.

Ten thousand parametric Poisson resamples of each measurement and its background were analyzed with fixed reference profiles. Cs-137 was top ranked in 100% of resamples at 70, 130, and 300 cm and in 99.79% at 390 cm. A background-only event gate based on the 99th percentile of the maximum NNLS coefficient was 0.028; every Cs-137 distance measurement exceeded this gate in the resampling analysis. This construction fixes the model-based background false-alarm probability at approximately 1%, but it does not estimate field false-alarm rates under changing environmental backgrounds.

At 390 cm the off-target Am-241 coefficient increased to 0.055 while the Cs-137 coefficient was 0.128. The correct source remained top ranked, but this secondary coefficient demonstrates that weak-signal multi-label interpretation requires radionuclide-specific thresholds or uncertainty bounds. Ranking performance should not be presented as proof that every secondary component is correctly rejected.

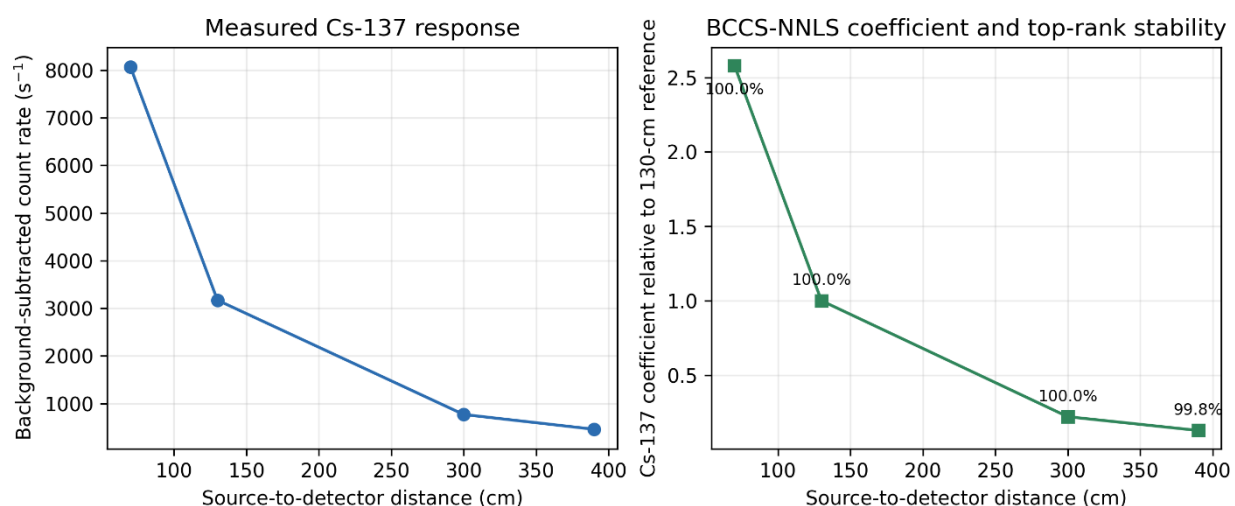


Figure 4. Distance-dependent Cs-137 response. The left panel shows the background-subtracted count rate over Ch. 4-256. The right panel shows the Cs-137 BCCS-NNLS coefficient relative to the 130-cm reference. Labels give the fraction of 10,000 parametric Poisson resamples in which Cs-137 was the top-ranked reference. The resampling quantifies counting-statistical stability only.

Table 3. BCCS-NNLS results for the Cs-137 distance series.

Distance (cm)	Net cps	Am-241	Cs-137	Co-60	Residual	Cs top-rank
70	8066.8	0.000	2.581	0.000	1.32%	100.00%
130	3170.9	0.000	1.000	0.000	0.00%	100.00%
300	769.7	0.026	0.220	0.014	1.17%	100.00%
390	460.1	0.055	0.128	0.000	6.39%	99.79%

Discussion

Characteristics and practical strengths of BCCS

The principal strength of BCCS is not that it creates new spectrometric information. It reorganizes the measured spectrum into a linear, peak-independent sequence that is convenient for constrained

decomposition. Each element represents counts above a cumulative threshold, so the method does not require a photopeak centroid, peak width, prominence criterion, or a small set of manually selected window edges.

Linearity is the clearest distinction from high-order energy weighting. For spectra C_1 and C_2 , $T(C_1 + C_2) = T(C_1) + T(C_2)$, where T denotes the BCCS transformation. Background subtraction is likewise consistent: $T(C - C_{bg}) = T(C) - T(C_{bg})$. Consequently, a mixed net spectrum can be modeled as a non-negative linear combination of single-radionuclide BCCS profiles. By contrast, $(C_1 + C_2)^2 E^3$ is not equal to $C_1^2 E^3 + C_2^2 E^3$; the higher-order energy-weighted transforms are useful for qualitative feature enhancement but not for direct linear attribution of mixture contributions.

BCCS also occupies a useful middle ground between raw-channel fitting and energy-window compression. Raw-spectrum NNLS retains the original channel-level information, whereas the five-window representation compresses the detector response into five integrated values. BCCS lies between these representations by retaining a cumulative coordinate for each selected threshold while integrating all counts above that threshold. This structure is well matched to plastic scintillators, where discriminatory information is distributed over broad continua.

The practical distinction of BCCS is therefore representational rather than informational. Unlike a small set of energy windows, BCCS retains a continuous sequence of detector-specific threshold responses. Unlike the higher-order energy-weighted transformations examined here, it preserves linear additivity and can therefore be combined directly with non-negative mixture decomposition. Unlike training-based classifiers, the present implementation requires reference spectra but no labelled training dataset. These properties make BCCS particularly suitable as an interpretable intermediate representation for broad, Compton-dominated plastic-scintillator responses.

The low-energy behavior is relevant to Am-241. High-order E^3 and E^6 weighting preferentially enhances higher channels and can reduce the relative visibility of a response concentrated near the LLD. BCCS does not impose such an energy exponent, and the Am-241 reference component remained among the two largest BCCS-NNLS coefficients in Sample-20. This should be interpreted as preservation of the low-channel Am-241 response in the fitted representation, not as measurement of a resolved 59.5-keV photopeak or as a validated detection decision.

Interpretation of coefficients and residuals

The NNLS coefficients quantify how strongly each fixed reference profile must be scaled to reconstruct the measured response. They are affected by source activity, distance, attenuation, detector efficiency, and acquisition geometry. They therefore support relative response attribution under fixed conditions but are not absolute activities. Activity estimation would require traceable source calibration and a response model that maps activity and geometry to expected counts. The low numerical BCCS residuals indicate close reconstruction within the cumulative representation space. For independent Poisson channel counts, the covariance between two cumulative thresholds is determined by the counts shared above the larger of the two thresholds. An unweighted BCCS-NNLS residual therefore should not be interpreted as a standard chi-square statistic or as independent evidence at every threshold. A future generalized least-squares implementation could include the full cumulative covariance matrix.

Under the known-K evaluation, BCCS-NNLS and five-window NNLS both placed the expected radionuclide pair among the two largest coefficients in all eight binary mixed-source measurements. Raw-spectrum NNLS produced the same rankings in the two detailed case studies and across all eight binary measurements. The energy-weighted representations qualitatively supported the same assignments for Samples 13 and 20, but their nonlinear form did not permit direct linear attribution of mixture components. Taken together, these comparisons show that BCCS-NNLS is technically feasible and interpretable under the present measurement conditions, but they do not establish general superiority over the alternative representations.

Practical scope and limitations

The present results demonstrate practical feasibility for static, unshielded sealed point-source measurements obtained with one large-area plastic-scintillator RPM under fixed gain, LLD, and background conditions. Separate vehicle-passage trials were also conducted at the testbed with and without radionuclide sources, and the system responses and radionuclide-identification outputs were inspected under operational conditions. However, these trials were not performed as a statistically designed validation campaign with complete archived spectra, repeated vehicle passages, predefined decision criteria, and traceable ground-truth records. They were therefore not included in the quantitative performance analysis presented in this study. Accordingly, performance under vehicle motion, intervening cargo, source shielding, detector-to-detector variation, and long-term environmental background changes remains to be quantitatively established.

The method requires stable channel gain, representative single-radionuclide references, and a background spectrum acquired under comparable conditions. Detector aging, temperature-dependent gain drift, changes in surrounding cargo, and background hardening can distort the reference relationship. BCCS reduces dependence on a small number of energy-window boundaries, but it is not invariant to gain drift or LLD changes.

The mixed-source evaluation included eight binary spectra and one ternary spectrum, with each condition represented by a single experimental acquisition. Samples 13 and 20 were selected only for detailed graphical presentation. Parametric resampling generated multiple counting-statistical realizations around the observed spectra, but it did not reproduce day-to-day system variation, independent source repositioning, vehicle-to-vehicle differences, or unmodeled background changes. The 100% top-two combination stability obtained for Samples 13 and 20 assumes that the number of constituents is known in advance and should not be interpreted as a general two-source identification rate.

A statistically powered validation should include repeated measurements for each single radionuclide, background-only condition, mixture, distance, shielding state, gain condition, and representative vehicle-passage scenario. It should predefine event and radionuclide decision thresholds and report confusion matrices, sensitivity, specificity, false-positive and false-negative rates, and performance over a controlled range of source-to-background ratios. Until such measurements are available, the present comparative results should be interpreted as case-study validation rather than statistical proof of performance superiority.

In conclusion, BCCS provides a linear and peak-independent representation of broad plastic-scintillator spectra that can be combined directly with NNLS decomposition. Under the present known-K, static point-source conditions, BCCS-NNLS recovered the expected constituent rankings in all eight binary mixtures and produced decomposition-consistent coefficients for the ternary mixture. Its practical distinction lies in

retaining a continuous sequence of cumulative threshold responses without requiring a training dataset or imposing high-order energy weighting. Although separate vehicle-passage trials provided preliminary operational observations, they were not included as part of the present quantitative validation. The correlated cumulative coordinates, limited independent repetitions, fixed reference library, and lack of statistically designed testing under representative cargo, shielding, and moving-vehicle conditions prevent claims of general performance superiority. Further validation using repeated measurements, predefined radionuclide decision thresholds, controlled gain and background variation, and documented portal-monitoring scenarios is required before open-set field deployment.

Methods

Detector system and acquisition settings

Measurements were performed at the cRPM-54 testbed at Gwangyang Port, Republic of Korea, using a plastic scintillation detector panel measuring 30 cm × 180 cm × 5 cm and coupled to a photomultiplier tube. The detector was operated with a custom measurement system developed by M1 International Co., Ltd. [13]. The system incorporated a photomultiplier high-voltage supply and data-acquisition electronics built around an Amptek DP5G digital pulse processor. The photomultiplier bias voltage was set to 1,475 V, and the DP5G gain setting was 4.0. Each background, single-radionuclide reference, and mixed-source pulse-height spectrum was acquired for 20 s. All spectra were converted to count rates using the recorded live time. Background subtraction was subsequently applied only to the analyses explicitly described as background-subtracted.

Source geometry and measurement configuration

All measurements were performed using sealed point sources. The source-to-detector distance was defined from the nominal center of each sealed source to the front surface of the detector panel. Each source was horizontally aligned with the central region of the detector and vertically positioned at the detector midpoint. For Samples 13–17, the constituent sources were placed adjacent to one another around the detector central axis at the same nominal source-to-detector distance using the same source holder. For Samples 18–21, the constituent sources were positioned at different source-to-detector distances, as listed in Supplementary Table S1.

The measurements were conducted under static, unshielded conditions without intervening cargo, intentional absorbers, or moving vehicles. Scattering from the detector housing, source holder, floor, and surrounding structures was not measured separately and was therefore implicitly included in the acquired reference and test spectra.

Channel scale, lower-level discriminator, and detector response

Although the DP5G supports a larger internal pulse-height scale, all spectra used in this study were exported and analyzed as 256 channels. Before the 20-s acquisitions, setup spectra of approximately 5–10 min were used to adjust the electronic gain so that the useful high-channel detector response remained within the exported 256-channel range. The photomultiplier bias voltage and gain setting were then fixed at 1475 V and 4.0, respectively, for all background, reference, and mixed-source measurements.

Channel 4 was the first accepted acquisition channel, whereas Channel 3 was below the lower-level discriminator setting. Channels 4–256 were therefore treated as the valid acquisition range. The detector produced a measurable response to Am-241, whose principal gamma emission is 59.5 keV. This observation indicates that the effective acquisition threshold was below 59.5 keV under the present operating conditions, but it does not establish a calibrated LLD energy. A calibrated threshold scan would be required to determine the LLD in physical energy units.

An absolute channel-to-energy calibration was not performed because the large plastic-scintillator detector produced broad, Compton-dominated pulse-height distributions without resolved full-energy peaks for the radionuclides examined. Consequently, a conventional photopeak-based energy resolution expressed as FWHM could not be meaningfully determined from these measurements. All BCCS, raw-spectrum, and window-based analyses were therefore performed in detector-channel space. The resulting channel boundaries are specific to the detector, gain, and LLD settings used in this study and should not be interpreted as universal energy intervals.

Table 4. Measurement and analysis configuration.

Item	Setting
Detector	cRPM-54 plastic-scintillator panel; 30 cm x 180 cm x 5 cm
Readout	Photomultiplier tube and Amptek DP5G digital pulse processor/MCA
Acquisition	256 exported channels; 20 s per spectrum; live-time corrected
Electronics	PMT high voltage 1,475 V; amplifier gain 4.0
Valid channels	Ch. 4-256; Ch. 3 below LLD
BCCS fit range	Cumulative thresholds Ch. 4-80
Background	121,966 counts over Ch. 4-256 in 20 s (6098.3 s ⁻¹)
Reference set	Am-241 Sample-1; Cs-137 Sample-6; Co-60 Sample-10
Detailed case-study mixtures	Sample-13: Cs-137 + Co-60; Sample-20: Am-241 + Cs-137

Reference and test spectra

The single-radionuclide reference set comprised Am-241 Sample-1 (90 µCi at 80 cm), Cs-137 Sample-6 (16 µCi at 130 cm), and Co-60 Sample-10 (9 µCi at 120 cm). These reference spectra were acquired using the same detector, high-voltage setting, amplifier gain, LLD setting, channel configuration, and 20-s acquisition time as the mixed-source spectra. The reference measurements were converted to live-time-normalized count rates and processed using the same background spectrum and analysis procedure as the corresponding test spectra.

An independently acquired 20-s background spectrum contained 121,966 counts over Channels 4–256, corresponding to 6098.3 s⁻¹. The same measured background spectrum was used for the primary background-subtracted analysis, the background-ratio visualization, and the parametric background resampling described below.

The Cs-137 distance series comprised measurements at 70, 130, 300, and 390 cm, with the 130-cm spectrum used as the fixed Cs-137 reference. All nine mixed-source spectra, Samples 13–21, were analyzed using BCCS-NNLS and five-window NNLS. Samples 13 and 20 were selected before graphical presentation as representative binary mixtures containing intermediate-to-high-energy and low-to-intermediate-energy responses, respectively. These two samples were also used for detailed raw-spectrum and energy-weighted comparisons.

Each reference, distance-series, and mixed-source condition was represented by a single experimental 20-s acquisition. The parametric resampling described below therefore quantifies counting-statistical variation around the observed spectra but does not replace independent repeated measurements.

The source activities and reference distances define the scale of the fitted NNLS coefficients. A coefficient of 1 indicates a detector response similar in magnitude to the corresponding background-corrected single-radionuclide reference response; it does not indicate equal radionuclide activity or activity fraction.

Backward cumulative channel sum

Let C_i be the count rate in channel i and let $N = 256$. The BCCS transform at threshold i is defined as

$$B_i = \sum_{j=i, \dots, N} C_j. \quad (1)$$

For an additive mixture consisting of background plus non-negative source components, application of the same linear operator gives

$$T[C(\text{mix}) - C(\text{bg})] = \sum_x a_x T(C_x). \quad (2)$$

BCCS was calculated over the full valid acquisition range, Channels 4–256. The NNLS measurement vector contained the cumulative threshold coordinates from Channels 4–80. Channels 4–80 define only the threshold positions entered into the fit; they do not restrict the counts included in each cumulative coordinate. Specifically, every BCCS value at threshold i contains all counts from Channel i through Channel 256. Therefore, counts in Channels 81–256, including the higher-channel portion of the Co-60 response, contribute to all BCCS coordinates used in the fit.

The Channel 4–80 threshold range was selected from the single-radionuclide reference profiles before evaluation of the mixed-source spectra because the largest relative differences among the reference BCCS curves occurred over these lower cumulative thresholds. The selected range corresponds to approximately 30% of the valid threshold coordinates and is specific to the present detector and gain configuration. It is not intended as a universal low-energy or low-channel definition.

Background compensation

The primary net BCCS for a measurement M was calculated after live-time normalization as

$$S_i = T(C^M / t^M)_i - T[C(\text{bg}) / t(\text{bg})]_i. \quad (3)$$

A background-ratio representation was also inspected:

$$R_i = T(C^M / t^M)_i / T[C(\text{bg}) / t(\text{bg})]_i - 1. \quad (4)$$

The subtraction-based representation preserves the net count-rate scale and was used for all primary BCCS-NNLS decompositions and quantitative residual calculations. Negative channel-level net count rates were

retained before the BCCS transformation to preserve exact linearity between the measured mixture and the reference components.

The ratio representation is dimensionless and emphasizes relative deviations from the background spectrum. However, it is sensitive to denominator uncertainty and to mismatch between the measured and representative background. It was therefore used only for visualization and was not used to calculate the NNLS coefficients, residuals, ranking statistics, or background event gate.

NNLS decomposition and residual

The processed measurement vector y was modeled using a reference matrix A whose columns were the background-subtracted BCCS profiles of Am-241, Cs-137, and Co-60. The coefficient vector a was obtained from

$$\hat{a} = \arg \min (a \geq 0) \|y - Aa\|_2^2. \quad (5)$$

The descriptive relative reconstruction residual was

$$r_{\text{rel}} = \|y - A\hat{a}\|_2 / \|y\|_2. \quad (6)$$

The fitted coefficients represent non-negative scaling factors for the fixed reference detector responses. They are influenced by source activity, distance, geometry, attenuation, and detector efficiency and were therefore not interpreted as radionuclide activity fractions.

For the eight binary mixtures, constituent recovery was evaluated under a known-K condition with $K=2$. A mixture was counted as recovered when its two expected radionuclides had the two largest fitted coefficients. This ranking criterion does not constitute an open-set multi-label classifier because the number of constituents was supplied in advance and no radionuclide-specific decision threshold was validated. For the ternary Sample-17, $K=3$ equals the total number of available reference radionuclides; consequently, top-K recovery is not discriminative, and only the fitted coefficients and reconstruction residual were reported as measures of decomposition consistency.

Because BCCS thresholds are correlated, the relative residual was used to describe reconstruction agreement rather than to calculate a chi-square probability.

Raw-spectrum NNLS baseline

As a comparison with the BCCS representation, the untransformed mixed-source spectrum was fitted over Channels 4–256 without prior subtraction of the background from the measurement vector. The raw-spectrum reference matrix contained the measured background template and the net single-radionuclide response profiles for Am-241, Cs-137, and Co-60. Each net radionuclide profile was calculated as the corresponding raw single-source reference spectrum minus the independently measured background spectrum after live-time normalization.

The raw mixed-source vector was modeled as

$$y_{\text{raw}} = \beta b + a_{\text{Am}} s_{\text{Am}} + a_{\text{Cs}} s_{\text{Cs}} + a_{\text{Co}} s_{\text{Co}} + \epsilon, \quad (7)$$

where b is the measured background template, s_x is the background-subtracted net response of radionuclide x , and all coefficients were constrained to be non-negative. The background coefficient β was excluded from the radionuclide ranking. The relative residual was calculated in the original channel space. This raw-

spectrum baseline was applied to all nine mixed-source spectra and to the four Cs-137 distance measurements using the same reference library and ranking criteria as BCCS-NNLS.

Five-window baseline

A detector-specific low-dimensional baseline was implemented using the energy-windowing concept of Ely et al. [3]. Five non-overlapping intervals were defined from the single-radionuclide reference responses before evaluation of the mixed-source spectra: Channels 4–10, 11–25, 26–40, 41–80, and 81–256. The window boundaries were then held fixed for all nine mixed-source measurements.

For each spectrum, counts were integrated within the five windows after live-time normalization. The raw five-window measurement vector was fitted using a background window template and the corresponding net single-radionuclide window responses. The high-channel interval, Channels 81–256, was included primarily to constrain the fitted background contribution, although source counts present in this interval were retained in the radionuclide reference vectors. The fitted background coefficient was excluded from the radionuclide ranking.

The same known-K ranking criterion and representation-specific relative residual were used for the five-window and BCCS analyses. Window-NNLS residuals were interpreted only within the five-dimensional window space and were not treated as directly comparable goodness-of-fit measures with the BCCS residuals.

Energy-weighted comparison

The energy-weighted comparison followed the three transformations reported by Lee et al. (2020) [6]:

$$W_{1,i} = C_i E_i; \quad W_{2,i} = C_i^2 E_i^3; \quad W_{3,i} = C_i^3 E_i^6. \quad (8)$$

Here, E_i denotes the normalized channel-proportional coordinate used in the numerical implementation. Specifically, $E_i = i/256$, where i is the MCA channel number. Because the spectra were analyzed without an absolute energy calibration, E_i should not be interpreted as photon energy in keV or MeV. Multiplication by a constant channel-to-energy scale would not change the peak positions of the individually normalized transformed spectra in channel space.

The detector-specific inspection intervals were fixed at Channels 4–10 for Am-241, 18–40 for Cs-137, and 41–90 for Co-60 based on the transformed single-radionuclide reference spectra. A reference-consistent local feature was considered present when its channel position fell within the corresponding interval. A radionuclide was qualitatively assigned when a consistent feature was observed in at least two of the three transformations.

After background subtraction, negative channel-level net values were set to zero before applying the nonlinear energy-weighted transformations. Each transformed spectrum was normalized by its maximum value. This comparison was used only for qualitative corroboration of Samples 13 and 20. The original energy-calibrated peak-window and count-ratio classification procedure of Lee et al. [6] was not reproduced in full, and the adapted implementation was not used as a quantitative open-set classifier.

Parametric counting-statistical resampling

Counting-statistical stability was examined using 10,000 parametric Poisson resamples with a fixed random seed. For each realization, every measurement and background channel was sampled independently using the observed channel count as the Poisson mean. Count rates were recalculated and fitted against the fixed measured reference matrix. A background-only event gate was defined as the 99th percentile of the maximum coefficient obtained when both pseudo-measurement and pseudo-background were sampled from the background spectrum; the resulting threshold was 0.028.

For each radionuclide coefficient, the resampling distribution was summarized using the median and the 2.5th and 97.5th percentiles, which were reported as a 95% counting-statistical interval. These intervals reflect Poisson variation in the observed measurement and background spectra while treating the experimentally measured reference matrix as fixed. They therefore do not include uncertainty caused by independent reference measurements, source repositioning, gain drift, detector aging, or environmental background changes.

For the single-source distance series, ranking stability was defined as the fraction of realizations in which Cs-137 had the largest radionuclide coefficient. For each binary mixed-source case, combination stability was defined as the fraction in which the two largest radionuclide coefficients matched the known pair. This latter calculation assumes that two sources are present and is not a general multi-label classification accuracy.

The 99th-percentile background-only gate used in this study is an empirical, model-based screening threshold and is not an ISO 11929 decision threshold or detection limit. A formal ISO 11929 evaluation would require a defined measurand, a complete uncertainty model, predefined error probabilities, and repeated background and source measurements under representative operating conditions.

Data availability

The raw MCA spectra, processed spectra, fitted coefficients, resampling results, and analysis tables supporting this study are available from Zenodo at [DOI]. The archived dataset includes the background, single-radionuclide reference, distance-series, and mixed-source measurements used in the analyses.

Code availability

The analysis code used for BCCS transformation, background correction, raw-spectrum NNLS, five-window NNLS, energy-weighted transformation, parametric Poisson resampling, and generation of the reported tables and figures is available from Zenodo at [DOI]. The archived repository includes the source code, dependency information, parameter settings, and instructions required to reproduce the analyses.

Acknowledgements

This research was supported by the Ministry of Oceans and Fisheries, Republic of Korea, through the Korea Institute of Marine Science & Technology Promotion (KIMST) under Project No. 20200611.

Author contributions

J.S. and **M.M.** contributed equally to the preparation and writing of the manuscript. **M.M.** conceived and supervised the study, performed the experiments, and served as the corresponding author. **J.S.** developed the analysis software and performed the signal processing. **B.J.** synthesized the test datasets and supported the evaluation. All authors contributed to the interpretation of the results and reviewed the manuscript.

Competing interests

The authors declare no competing interests.

References

1. Siciliano, E. et al. Comparison of PVT and NaI(Tl) scintillators for vehicle portal monitor applications. *Nucl. Instrum. Methods Phys. Res. A* 550, 647-674 (2005).
2. Bertrand, G. H. V., Hamel, M. & Sguerra, F. Current status on plastic scintillator modifications. *Chem. Eur. J.* 20, 15660-15685 (2014).
3. Ely, J., Kouzes, R., Schweppe, J., Siciliano, E., Strachan, D. & Weier, D. The use of energy windowing to discriminate SNM from NORM in radiation portal monitors. *Nucl. Instrum. Methods Phys. Res. A* 560, 373-387 (2006).
4. Hevener, R., Yim, M.-S. & Baird, K. Investigation of energy windowing algorithms for effective cargo screening with radiation portal monitors. *Radiat. Meas.* 58, 113-120 (2013).
5. Lee, H. C. et al. Radioisotope identification using an energy-weighted algorithm with a proof-of-principle radiation portal monitor based on plastic scintillators. *Appl. Radiat. Isot.* 156, 109010 (2020).
6. Lee, H. C. et al. Evaluation of source identification method based on energy-weighting level with portal monitoring system using plastic scintillator. *J. Radiat. Prot. Res.* 45, 117-129 (2020).
7. Paff, M. G., Di Fulvio, A., Clarke, S. D. & Pozzi, S. A. Radionuclide identification algorithm for organic scintillator-based radiation portal monitor. *Nucl. Instrum. Methods Phys. Res. A* 849, 41-48 (2017).
8. Runkle, R., Tardiff, M., Anderson, K., Carlson, D. & Smith, L. Analysis of spectroscopic radiation portal monitor data using principal components analysis. *IEEE Trans. Nucl. Sci.* 53, 1418-1423 (2006).
9. Kim, Y., Kim, M., Lim, K. T., Kim, J. & Cho, G. Inverse calibration matrix algorithm for radiation detection portal monitors. *Radiat. Phys. Chem.* 155, 127-132 (2019).
10. Kim, J., Park, K. & Cho, G. Multi-radioisotope identification algorithm using an artificial neural network for plastic gamma spectra. *Appl. Radiat. Isot.* 147, 83-90 (2019).
11. Jeon, B., Kim, J., Yu, Y. & Moon, M. Comparison of machine learning-based radioisotope identifiers for plastic scintillation detector. *J. Radiat. Prot. Res.* 46, 204-212 (2021).
12. Lawson, C. L. & Hanson, R. J. *Solving Least Squares Problems*. (Society for Industrial and Applied Mathematics, 1995).
13. M1 International Co., Ltd. Container-interior radiation detection system. <https://m1int1.cafe24.com/article/business/8/24/> (2022) (in Korean).

Supplementary Table S1

Measurement conditions for the background, single-source, and mixed-source spectra.

Data ID	Measurement category	Radionuclide(s)	Nominal activity (μCi)	Distance (cm)	Role in the analysis
Background	Background-only	-	-	-	Background correction, normalization, and background-only gate
Sample-1	Single-source	Am-241	90	80	Am-241 reference spectrum
Sample-2	Single-source	Am-241	90	180	Single-source distance measurement
Sample-3	Single-source	Am-241	90	280	Single-source distance measurement
Sample-4	Single-source	Am-241	90	460	Single-source distance measurement
Sample-5	Single-source	Cs-137	16	70	Cs-137 distance series
Sample-6	Single-source	Cs-137	16	130	Cs-137 reference spectrum
Sample-7	Single-source	Cs-137	16	300	Cs-137 distance series
Sample-8	Single-source	Cs-137	16	390	Cs-137 distance series
Sample-9	Single-source	Co-60	9	60	Single-source distance measurement
Sample-10	Single-source	Co-60	9	120	Co-60 reference spectrum
Sample-11	Single-source	Co-60	9	310	Single-source distance measurement
Sample-12	Single-source	Co-60	9	470	Single-source distance measurement
Sample-13	Mixed-source	Cs-137 Co-60	Cs-137: 16 Co-60: 9	Cs-137: 136 Co-60: 136	Mixed-source dataset; cross-method case study
Sample-14	Mixed-source	Am-241 Co-60	Am-241: 90 Co-60: 9	Am-241: 287 Co-60: 287	Mixed-source dataset
Sample-15	Mixed-source	Cs-137 Co-60	Cs-137: 16 Co-60: 9	Cs-137: 361 Co-60: 361	Mixed-source dataset
Sample-16	Mixed-source	Am-241 Cs-137	Am-241: 90 Cs-137: 16	Am-241: 163 Cs-137: 163	Mixed-source dataset
Sample-17	Mixed-source	Am-241 Cs-137 Co-60	Am-241: 90 Cs-137: 16 Co-60: 9	Am-241: 309 Cs-137: 309 Co-60: 309	Mixed-source dataset
Sample-18	Mixed-source	Co-60 Am-241	Co-60: 9 Am-241: 90	Co-60: 153 Am-241: 210	Mixed-source dataset; separated source positions
Sample-19	Mixed-source	Cs-137 Co-60	Cs-137: 16 Co-60: 9	Cs-137: 367 Co-60: 285	Mixed-source dataset; separated source positions
Sample-20	Mixed-source	Am-241 Cs-137	Am-241: 90 Cs-137: 16	Am-241: 85 Cs-137: 173	Mixed-source dataset; separated positions; cross-method case study
Sample-21	Mixed-source	Cs-137 Co-60	Cs-137: 16 Co-60: 9	Cs-137: 223 Co-60: 316	Mixed-source dataset; separated source positions

Source-to-detector distance was measured from the nominal center of each sealed point source to the front surface of the detector panel. All sources were aligned with the central region of the detector, and their vertical positions corresponded to the detector midpoint. “Equal-distance” indicates that all constituent sources were placed at the same nominal distance, whereas “unequal-distance” indicates that different source-to-detector distances were used for the constituent sources.

Notes. Activities are the nominal source activities reported in the experimental-condition sheet. For mixed-source measurements, radionuclides, activities, and distances are listed in corresponding order. For Samples 13-17, all constituent radionuclides have the same reported distance; for Samples 18-21, the constituent sources have different reported positions.

All spectra were acquired for 20 s using 256 exported MCA channels at a PMT high voltage of 1,475 V and an amplifier gain of 4.0; Channels 4-256 were treated as valid acquisition channels. The background spectrum contained 121,966 counts over Channels 4-256 in 20 s (6,098.3s⁻¹). Distances are the physical reference point from the detector surface.

Supplementary Table S2

Results of BCCS-NNLS and five-window NNLS for all mixed-source spectra under the known-constituent-number evaluation

Sample	Expected radionuclides	Expected constituents recovered in top (K), BCCS-NNLS	BCCS residual (%)	Expected constituents recovered in top (K), five-window NNLS	Five-window residual (%)
Sample-13	Cs-137 + Co-60	2/2	0.37	2/2	1.05
Sample-14	Am-241 + Co-60	2/2	1.17	2/2	2.19
Sample-15	Cs-137 + Co-60	2/2	0.89	2/2	1.99
Sample-16	Am-241 + Cs-137	2/2	0.50	2/2	1.45
Sample-17	Am-241 + Cs-137 + Co-60	N/A	1.24	N/A*	2.55
Sample-18	Am-241 + Co-60	2/2	0.39	2/2	1.18
Sample-19	Cs-137 + Co-60	2/2	0.76	2/2	2.07
Sample-20	Am-241 + Cs-137	2/2	0.77	2/2	0.77
Sample-21	Cs-137 + Co-60	2/2	1.03	2/2	1.66

Notes. K denotes the known number of constituent radionuclides in each mixed-source spectrum. Recovery was considered successful when all expected radionuclides were included among the K largest NNLS coefficients. Therefore, this evaluation represents known-K constituent ranking rather than open-set radionuclide identification. Relative residuals were calculated separately in the BCCS and five-window representation spaces and should not be interpreted as directly comparable goodness-of-fit measures between the two methods. Each sample represents a single 20-s experimental acquisition.

*The top-K ranking metric is not discriminative for Sample-17 because K=3 equals the number of candidate reference radionuclides.

Supplementary Table S3

Raw-spectrum NNLS results for all mixed-source spectra using a measured background template

Sample	Expected radionuclides	Am-241 coefficient	Cs-137 coefficient	Co-60 coefficient	Background coefficient	Residual	Expected constituents recovered in top K
Sample-13	Cs-137 + Co-60						

Sample-14	Am-241 + Co-60						
Sample-15	Cs-137 + Co-60						
Sample-16	Am-241 + Cs-137						
Sample-17	Am-241 + Cs-137 + Co-60						N/A
Sample-18	Am-241 + Co-60						
Sample-19	Cs-137 + Co-60						
Sample-20	Am-241 + Cs-137						
Sample-21	Cs-137 + Co-60						

Notes. Raw-spectrum NNLS was applied to the untransformed mixed-source spectra over Channels 4–256 without prior subtraction of the background from the measurement vector. The reference matrix contained the measured background template and the background-subtracted net response profiles of Am-241, Cs-137, and Co-60. The fitted background coefficient was excluded from radionuclide ranking. K denotes the known number of radionuclide constituents in each mixed-source measurement. Recovery was considered successful when all expected radionuclides were included among the K largest radionuclide coefficients. The top- K metric is not discriminative for Sample-17 because $K=3$ equals the number of candidate radionuclides. Relative residuals were calculated in the original channel space and should not be directly compared with residuals calculated in the BCCS or five-window representation spaces.

Supplementary Table S4

Parametric resampling intervals for BCCS-NNLS coefficients in the representative mixed-source spectra

Sample	Radionuclide	Fitted coefficient	Resampling median	95% counting statistical interval
Sample-13	Am-241	0.031		
Sample-13	Cs-137	0.951		
Sample-13	Co-60	0.796		
Sample-20	Am-241	0.946		
Sample-20	Cs-137			
Sample-20	Co-60			

Notes. The fitted coefficient is the value obtained from the experimentally measured spectrum. The resampling median and 95% counting-statistical interval were calculated from 10,000 parametric Poisson realizations using the 2.5th and 97.5th percentiles of the coefficient distribution. The experimentally measured reference matrix was held fixed during resampling. Therefore, the intervals represent counting-statistical variation in the measured mixed-source and background spectra and do not include uncertainty arising from independent reference measurements, source positioning, detector gain drift, detector aging, or environmental background changes.

